

Notes: Gulen & Ion (RFS, 2016)

Policy Uncertainty and Corporate Investment

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1 Big picture

- **Question.** Does uncertainty about future government policy and regulation depress corporate capital investment?
- **Core hypothesis.** Policy uncertainty lowers investment because many investment projects are partly irreversible. When the option value of waiting rises, firms delay capital expenditures.
- **Main empirical object.** The paper studies firm-quarter capital expenditure scaled by beginning-of-period total assets, and relates it to the Baker–Bloom–Davis policy uncertainty index.
- **Baseline result.** Policy uncertainty is negatively associated with future firm investment. The relation remains when the authors control for Tobin’s q , cash flow, sales growth, GDP growth, election years, and seasonality.
- **Magnitude.** In the preferred specifications with controls for investment opportunities, a doubling of policy uncertainty is associated with a decline in next-quarter investment equal to roughly 8.7% of the sample mean.
- **Cross-sectional mechanism.** The negative effect is much stronger when investment is more irreversible: firms with high PPE, low asset redeployability, high sunk costs, and durable-goods exposure are more sensitive.
- **Government channel.** The effect is also stronger in industries more dependent on government spending, consistent with policy uncertainty mattering more when future demand is shaped by public budgets.
- **Dynamic pattern.** Investment declines after policy uncertainty rises, then rebounds later. The rebound is especially visible for firms with more reversible assets.
- **Identification concern.** Policy uncertainty may proxy for bad investment opportunities or general macroeconomic uncertainty. The paper addresses this with extensive controls, out-of-sample tests, a Canadian-policy-uncertainty residual strategy, and a political-polarization instrument.
- **Takeaway.** Policy uncertainty is not merely a macro narrative variable. It predicts firm-level investment in ways aligned with real-options theory: the effect is strongest where delaying investment is most valuable.

2 Incremental contribution

- **From uncertainty to policy uncertainty.** Earlier investment-under-uncertainty work often studies general volatility. This paper isolates uncertainty associated specifically with future policy and regulation.
- **From aggregate correlations to firm-level investment.** Instead of only showing that policy uncertainty spikes in recessions, the paper links the policy uncertainty index to firm-quarter capital expenditures for hundreds of thousands of observations.
- **Mechanism through irreversibility.** The central contribution is not just documenting a negative average relation. The paper shows that the relation is systematically stronger when firms’ investment is harder to reverse.
- **Government exposure as a second channel.** By showing stronger effects for firms more sensitive to government spending, the paper connects policy uncertainty to firms’ expected demand environment.
- **Endogeneity mitigation.** The Canadian residual and political-polarization IV strategies extend beyond standard controls, directly addressing the concern that the index simply captures recessionary macro conditions.
- **Dynamic evidence.** The paper documents investment delay and later rebound after high-policy-uncertainty spells, making the mechanism closer to real-options timing than to a permanent decline in investment opportunities.

- **Contribution to corporate finance.** The paper turns policy uncertainty into a corporate investment determinant, comparable in empirical importance to standard variables such as Tobin’s q , cash flow, and sales growth.

3 Data and measurement

3.1 Policy uncertainty

- The main measure is the Baker, Bloom, and Davis news-based policy uncertainty index.
- The index counts newspaper articles that jointly mention uncertainty, economic conditions, and policy-related terms.
- The paper also studies subcomponents related to news, tax policy, and government purchases/inflation.
- For firm-quarter regressions, policy uncertainty is the logarithm of the average monthly index value over the three months of a firm’s fiscal quarter.

Interpretation: The use of fiscal-quarter alignment gives mild cross-sectional variation in policy uncertainty, but the index is economically close to an aggregate time-series variable.

3.2 Firm investment variables

- The dependent variable is capital expenditures over beginning-of-period total assets, denoted $CAPX/TA$.
- Tobin’s q is measured as market equity plus book assets minus book equity plus deferred taxes, divided by book assets.
- Cash flow is operating cash flow over beginning-of-period assets.
- Sales growth is year-on-year quarterly sales growth.
- The final sample contains 441,326 firm-quarter observations in the baseline summary statistics.

Interpretation: Scaling investment and cash flow by assets makes coefficients comparable across firms of different sizes. The specification remains a panel regression, not an aggregate time-series regression.

3.3 Investment irreversibility proxies

- **PPE intensity:** firms with more property, plant, and equipment have more physical, long-lived assets.
- **Asset redeployability:** lower redeployability means assets have fewer alternative users and are harder to sell without a discount.
- **Sunk-cost index:** high sunk costs make investment less reversible.
- **Durables exposure:** durable-goods industries tend to have investment demand more exposed to policy and macro uncertainty.

Interpretation: These proxies allow the paper to test the core real-options prediction: uncertainty should matter more when reversing the investment is costly.

3.4 Government-spending exposure

- The paper constructs a measure of industry sensitivity to government spending.
- Firms in industries more exposed to government demand should be more affected by policy uncertainty because future revenues depend more directly on policy decisions.

Interpretation: This is a second cross-sectional mechanism. It focuses on policy uncertainty as uncertainty about demand and budgetary decisions, not just uncertainty about regulation.

4 Models and empirical specifications

4.1 Baseline investment regression

The baseline panel regression is

$$\frac{CAPX_{i,t+l}}{TA_{i,t+l-1}} = \alpha_i + \beta_1 PU_{i,t} + \beta_2 TQ_{i,t} + \beta_3 \frac{CF_{i,t}}{TA_{i,t-1}} + \beta_4 SG_{i,t} + \delta M_t + QRT_t + \varepsilon_{i,t+l}. \quad (1)$$

Here $l \in \{1, 2, 3, 4\}$ indexes future quarters; α_i are firm fixed effects; QRT_t includes calendar- and fiscal-quarter dummies; and M_t contains macro controls.

Interpretation: The coefficient β_1 measures whether higher policy uncertainty today predicts lower capital investment over the next one to four quarters, after controlling for standard investment determinants.

4.2 Controls for investment opportunities and general uncertainty

The authors add proxies for omitted investment opportunities and economic uncertainty:

- expected GDP growth;
- the Conference Board leading economic index;
- consumer confidence;
- GDP forecast dispersion;
- profit-growth volatility;
- VXO implied volatility;
- cross-sectional return volatility;
- the Jurado–Ludvigson–Ng macro uncertainty measure.

Interpretation: This is important because policy uncertainty spikes during recessions, elections, wars, and financial crises. The paper tries to show that policy uncertainty has predictive content beyond general bad times.

4.3 Out-of-sample predictive test

The paper compares models with and without policy uncertainty using the Clark–West adjusted MSPE statistic:

$$Adj.\Delta MSPE_l^j = \frac{2}{N} \sum_{i,t} PE_{i,t,l}^j \left(PE_{i,t,l}^j - PE_{i,t,l}^{j,PU} \right). \quad (2)$$

Interpretation: Positive and significant values mean that adding policy uncertainty improves out-of-sample forecasts of firm investment relative to benchmark models.

4.4 Cross-sectional heterogeneity regression

The mechanism tests interact policy uncertainty with irreversibility or government exposure:

$$\frac{CAPX_{i,t+l}}{TA_{i,t+l-1}} = \alpha_i + \gamma_t + \beta_1 PU_{i,t} \cdot H_{i,t} + \beta_2 H_{i,t} + \beta_3 TQ_{i,t} + \beta_4 \frac{CF_{i,t}}{TA_{i,t-1}} + \beta_5 SG_{i,t} + \varepsilon_{i,t+l}. \quad (3)$$

Interpretation: If $H_{i,t}$ captures irreversibility, then a negative β_1 means policy uncertainty is more damaging for firms whose investment is harder to reverse.

4.5 VAR impulse response

The aggregate VAR includes policy uncertainty, consumer confidence, and aggregate versions of q , cash flow, sales growth, and investment.

Interpretation: The VAR turns the panel results into a dynamic impulse-response exercise and asks how aggregate investment reacts after a policy uncertainty shock.

4.6 Canadian residual policy uncertainty

The residual strategy first estimates a time-series regression of U.S. policy uncertainty on Canadian policy uncertainty and controls:

$$USPU_t = \alpha + \beta_1 CANPU_t + \beta_2 TQ_t + \beta_3 CF_t + \beta_4 SG_t + \theta INT_t + \delta M_t + QRT_t + \varepsilon_t. \quad (4)$$

The residual becomes RPU_t , an orthogonalized U.S. policy uncertainty measure.

Interpretation: Canadian policy uncertainty is used to purge macroeconomic uncertainty common across both countries. If the residual still predicts U.S. investment but not Canadian investment, the result is less likely to be a generic macro shock.

4.7 Political-polarization instrument

The IV strategy uses U.S. Senate political polarization as an instrument for policy uncertainty:

$$PU_t = \alpha + \beta_1 POLAR_t + \beta_2 TQ_t + \beta_3 CF_t + \beta_4 SG_t + \theta INT_t + \delta M_t + QRT_t + \varepsilon_t. \quad (5)$$

Interpretation: Political polarization is meant to shift policy uncertainty by making policy decisions more difficult and less predictable, while being less directly tied to firms' immediate investment opportunities.

5 Identification and interpretation

5.1 What identifies the average effect?

- The baseline evidence uses time-series variation in national policy uncertainty combined with firm-level panel investment data.

- Firm fixed effects absorb time-invariant firm investment propensities.
- Quarter controls absorb seasonality.
- Macroeconomic controls absorb time variation in investment opportunities and general uncertainty.

Interpretation: The baseline relation is not a randomized shock. Its credibility comes from robustness, timing, out-of-sample predictability, and cross-sectional patterns that match theory.

5.2 Why the heterogeneity evidence is powerful

- If policy uncertainty only proxies for recessions, the effect need not be systematically stronger for high-irreversibility firms.
- The real-options mechanism predicts exactly this pattern.
- Government-spending exposure provides a separate channel tied specifically to policy demand.

Interpretation: The heterogeneity tests make the paper more than an aggregate correlation study. They test a mechanism.

5.3 What remains hard to prove

- Policy uncertainty is highly correlated with crises and weak macro conditions.
- The index is mostly aggregate, limiting the use of time fixed effects.
- Firms may also change investment because expected demand falls, not only because uncertainty rises.
- Irreversibility proxies are imperfect and may capture other industry characteristics.
- Political polarization may affect firms through channels other than uncertainty.

Interpretation: The paper mitigates these concerns but cannot fully eliminate them. Its main claim is best read as a strong empirical pattern with theory-consistent cross-sectional and dynamic evidence.

6 Tables and figures

6.1 Figure 1: Policy uncertainty index + Table 1: Summary statistics

Read:

- Figure 1 plots the Baker–Bloom–Davis policy uncertainty index from 1987 to 2013 and marks major events such as the Gulf Wars, 9/11, the financial crisis, debt-ceiling disputes, the Euro debt crisis, and the government shutdown.
- The index rises around events that plausibly increase uncertainty about taxes, spending, regulation, and macro policy.
- Table 1 shows that the overall policy uncertainty index is highly correlated with its news component and negatively related to real GDP growth.
- The sample has 441,326 firm-quarter observations, with mean capital expenditures equal to 1.4% of lagged assets.
- The classic investment regression in Panel C confirms that both Tobin's q and cash flow predict investment.

Economic message: Figure 1 motivates policy uncertainty as a time-varying state variable. Table 1 establishes the firm-level investment panel and shows why controls for investment opportunities matter.

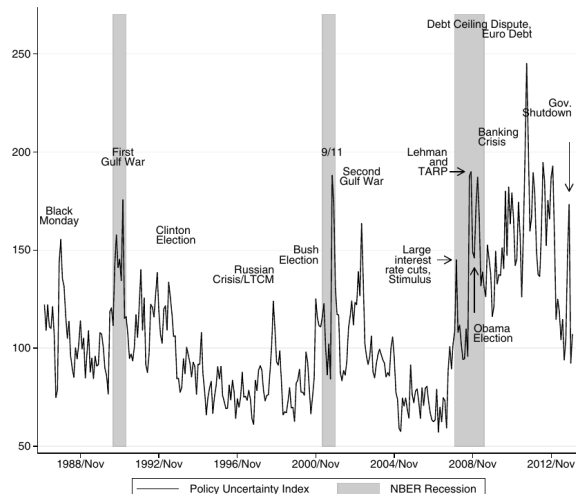


Figure 1: Policy uncertainty index

Table 1
Summary statistics

Panel A: Correlation matrix

	PU	PU news	PU Tax	PU CPI	PU GOV	Δ GDP	VXO
Policy uncertainty (PU)	1.000						
PU news component	0.887	1.000					
PU tax component	0.626	0.357	1.000				
PU CPI component	0.471	0.153	0.258	1.000			
PU Gov.Purch. component	0.373	0.071	0.156	0.427	1.000		
Real GDP growth	-0.396	-0.393	-0.182	-0.270	0.009	1.000	
VXO	0.383	0.457	0.017	0.210	-0.025	-0.276	1.000

Panel B1: Sample used in this study

Panel B2: Compustat universe 1987–2013

	N	Mean	Median	SD	N	Mean	Median	SD
CAPX	441,326	26.5	1.5	87.4	572,204	24	1.1	83.5
PPE	425,108	648	34.2	2,123	639,932	609	24.3	2,075
Total assets	441,326	2,882	267	9,170	661,360	2,886	226	9,418
Operating cash flows	441,326	53.5	2.8	190	544,309	48.7	1.9	183
Sales	441,326	423	48.4	1,187	649,950	397	36.6	1,185
CAPX/Lag total assets	441,326	0.014	0.008	0.020	568,769	0.015	0.008	0.022
Tobin's q	441,326	1.869	1.350	1.545	643,699	1.937	1.319	1.733
Cash flows/Lag total assets	441,326	0.011	0.015	0.058	541,294	0.006	0.013	0.065
Sales growth	441,326	0.181	0.078	0.581	588,911	0.192	0.080	0.617
PPE/Lag total assets	425,108	0.265	0.187	0.245	617,089	0.259	0.172	0.254

Panel C: Classic investment regressions

	CAPX/TA	CAPX/TA	CAPX/TA
Tobin's q	0.156*** (24.66)		0.157*** (24.85)
Cash Flows		0.0517*** (14.98)	0.0533*** (16.26)
N	441,326	441,326	441,326
R-sq	0.020	0.003	0.023

This table presents summary statistics for the main variables used in our analysis. The data extend from January 1987 to December 2013. Panel A presents correlations between the policy uncertainty index of Baker, Bloom,

Table 1: Summary statistics

6.2 Table 2: Policy uncertainty and capital investment, Panels A–B + Table 2: Panels C–D

Read:

- Panel A shows that the overall policy uncertainty index predicts lower investment one to four quarters ahead.
- Panel B shows that the news-based policy uncertainty component produces similar negative coefficients.
- Panel C shows that the tax-code component is also negatively related to investment.
- Panel D, based on government spending and inflation news, is weaker and often statistically insignificant.

- The coefficients remain negative with firm fixed effects, quarter dummies, and controls for Tobin's q , cash flow, sales growth, GDP growth, and elections.

Economic message: The baseline result is broad: policy uncertainty predicts lower capital investment, especially when measured by the overall index or the news component.

Dependent variable: CAPX/Total assets	Panel A: Overall policy uncertainty index				Panel B: News-based component of policy uncertainty index			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Policy uncertainty	-0.166*** (-6.67)	-0.171*** (-6.75)	-0.179*** (-7.04)	-0.174*** (-6.88)	-0.126*** (-5.48)	-0.137*** (-5.88)	-0.161*** (-6.32)	-0.153*** (-6.54)
Tobin's q	0.169*** (24.43)	0.159*** (23.63)	0.143*** (22.18)	0.128*** (18.65)	0.171*** (24.49)	0.169*** (23.60)	0.143*** (22.15)	0.128*** (18.59)
Cash flow	0.0258*** (10.06)	0.0359*** (13.70)	0.0379*** (14.22)	0.0347*** (13.31)	0.0257*** (9.95)	0.0357*** (13.59)	0.0378*** (14.13)	0.0346*** (13.31)
Sales growth	0.049*** (14.69)	0.0448*** (15.52)	0.0397*** (13.84)	0.0305*** (10.54)	0.0414*** (14.35)	0.0452*** (15.41)	0.0406*** (13.90)	0.0397*** (10.62)
GDP growth	0.0111 (1.43)	0.0109** (2.11)	0.0225** (2.40)	0.0324*** (3.39)	0.0137* (1.70)	0.0219** (2.55)	0.0241*** (3.40)	0.0315*** (4.40)
Election indicator	0.00448 (0.28)	-0.00517 (-0.33)	-0.0155 (-1.23)	-0.0251 (-2.01)	0.00674 (0.41)	-0.00291 (-0.18)	-0.0131 (-1.07)	-0.0227 (-1.14)
N	424,785	412,621	401,744	392,679	424,785	412,621	401,744	392,679
R-squared	0.09	0.08	0.03	0.08	0.08	0.037	0.03	0.08
Firm fixed effects	yes	yes	yes	yes	yes	yes	yes	yes
Quarter dummies	yes	yes	yes	yes	yes	yes	yes	yes
Cluster by firm	yes	yes	yes	yes	yes	yes	yes	yes
Cluster by quarter	yes	yes	yes	yes	yes	yes	yes	yes

(continued)

Dependent variable: CAPX/Total assets	Panel C: Policy uncertainty related to tax code				Panel D: Policy uncertainty related to government spending and inflation			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Policy uncertainty	-0.0601*** (-14.12)	-0.0565*** (-13.10)	-0.0536*** (-11.78)	-0.0510*** (-10.54)	-0.0331 (-0.77)	-0.0336 (-0.77)	-0.0381 (-0.62)	-0.0358 (-0.20)
Tobin's q	0.166*** (23.91)	0.156*** (23.17)	0.142*** (21.48)	0.127*** (18.22)	0.160*** (24.84)	0.152*** (24.07)	0.152*** (22.51)	0.134*** (19.33)
Cash flow	0.0269*** (10.65)	0.0370*** (14.39)	0.0391*** (14.88)	0.0256*** (9.91)	0.0269*** (10.65)	0.0256*** (9.91)	0.0270*** (10.65)	0.0246*** (9.91)
Sales growth	0.0842*** (13.41)	0.0425*** (14.63)	0.0279*** (12.95)	0.0269*** (9.77)	0.0425*** (14.63)	0.0279*** (12.95)	0.0269*** (9.77)	0.0269*** (9.77)
GDP growth	0.0146*** (2.42)	0.0238*** (3.51)	0.0276*** (3.75)	0.0274*** (4.55)	0.0276*** (3.75)	0.0274*** (4.55)	0.0276*** (3.75)	0.0276*** (3.75)
Election indicator	0.00391 (0.29)	-0.00491 (-0.34)	-0.0044 (-0.34)	-0.0044 (-1.12)	0.00927 (0.56)	-0.00173 (-0.01)	-0.00173 (-0.01)	-0.00173 (-0.01)
N	424,785	412,621	401,744	392,679	424,785	412,621	401,744	392,679
R-squared	0.046	0.044	0.039	0.032	0.046	0.044	0.039	0.032
Firm fixed effects	yes	yes	yes	yes	yes	yes	yes	yes
Quarter dummies	yes	yes	yes	yes	yes	yes	yes	yes
Cluster by firm	yes	yes	yes	yes	yes	yes	yes	yes
Cluster by quarter	yes	yes	yes	yes	yes	yes	yes	yes

In this table we regress firm-level quarterly investment (CAPX/Lagged Total Assets) on Tobin's q , operating cash flows, sales growth, and the policy uncertainty index from Baker, Bloom, and Davis (2013) (panel A). In panels B through D, we replace the overall policy uncertainty index with each of its three components, respectively. The data are quarterly and extend from January 1987 to December 2013. See Section 2 for a detailed description of how we calculate each variable. In specifications marked (1), the dependent variable has a lead of one period (calendar quarter) with respect to the independent variables in specifications marked (2) it leads two periods, and so forth unit (4). All specifications include calendar and fiscal quarter dummies, as well as firm fixed effects. All variables are normalized by their sample standard deviation. Standard errors are clustered at the quarter and firm-level. t -statistics are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

Table 2: Policy uncertainty and capital investment, Panels A–B

Table 2: Policy uncertainty and capital investment, Panels C–D

6.3 Table 3: Alternative macroeconomic controls + Table 4: Out-of-sample predictability

Read:

- Table 3 shows that policy uncertainty remains negative after adding controls for expected GDP growth, leading indicators, consumer confidence, forecast dispersion, VXO, return dispersion, and macro uncertainty.
- The coefficient magnitude falls when richer controls are added, but the relation remains statistically meaningful.
- Table 4 shows that adding policy uncertainty improves out-of-sample predictions relative to benchmark models.
- The forecasting exercise matters because it asks whether policy uncertainty contains information useful for predicting future investment rather than merely explaining investment in-sample.

Economic message: Policy uncertainty has predictive content beyond standard macro controls. The out-of-sample evidence reduces concerns that the result is purely overfitting or an in-sample correlation.

Table 3
Alternative macroeconomic controls for investment opportunities and economic uncertainty

	Panel A: Add controls for investment opportunities				Panel B: Add controls for economic uncertainty			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Policy uncertainty (news)	-0.0007** (-2.15)	-0.0019** (-2.55)	-0.0185*** (-2.85)	-0.0602*** (-3.05)	-0.0909*** (-3.11)	-0.0963*** (-3.56)	-0.1107*** (-3.79)	-0.1227*** (-4.04)
GDP growth	0.00193 (0.49)	0.00843 (1.00)	0.00843 (0.97)	0.0175* (1.91)	0.00275 (0.43)	0.00677 (0.98)	0.00686 (0.93)	0.0156* (1.83)
Expected GDP growth	0.00837 (1.26)	0.0131* (1.96)	0.0152** (2.16)	0.0161** (2.22)	0.000461 (0.08)	0.00507 (0.83)	0.00644 (1.04)	0.00591 (0.93)
Leading economic index	-0.00326 (-0.43)	0.00799 (1.08)	0.0156** (2.53)	0.0205*** (3.22)	-0.0553*** (-5.01)	-0.0415*** (-3.84)	-0.0339*** (-2.90)	-0.0283*** (-2.69)
Consumer confidence	0.00318*** (4.25)	0.00263*** (3.99)	0.00224*** (3.28)	0.00162*** (2.21)	0.00105 (1.48)	0.000590 (0.86)	0.000265 (0.29)	-0.000421 (-0.52)
GDP forecast dispersion					-0.00710 (-0.85)	-0.00835 (-1.05)	-0.00855 (-1.05)	-0.00992 (-1.52)
Profit growth SD					-0.0566** (-2.49)	-0.0335** (-2.45)	-0.0224* (-1.71)	-0.0111 (-0.79)
VXO					0.0211** (2.39)	0.0194** (2.33)	0.0168* (1.72)	0.0113 (1.20)
Returns SD					0.0163** (2.30)	0.0169** (2.42)	0.0179*** (2.36)	0.0228*** (1.45)
JLN uncertainty measure					-0.0727*** (-3.97)	-0.0700*** (-4.19)	-0.0703*** (-4.77)	-0.0661*** (-5.45)
Controls	yes	yes	yes	yes	yes	yes	yes	yes
N	424,765	412,621	401,744	392,679	416,118	409,803	401,744	392,679
R-squared	0.040	0.039	0.035	0.030	0.044	0.043	0.039	0.034

In this table we present results obtained from estimating our baseline investment equation using several alternative macroeconomic proxies for investment opportunities (panel A) and economic uncertainty (panel B). The proxies for investment opportunities are the realized net GDP growth, the expected GDP growth calculated biannually from the Livingston survey of the Philadelphia Federal Reserve Bank, the Leading Economic Index released by The Conference Board, and the Michigan Consumer Confidence Index developed by the University of Michigan. The proxies for economic uncertainty are the coefficient of variation of the biannual GDP forecasts from the Livingston survey of the Philadelphia Federal Reserve Bank, the cross-sectional standard deviation in firm-level profit growth, the monthly VXO implied volatility index from the CBOE, the cross-sectional standard deviation in firm-level monthly stock returns, and the comprehensive measure of macroeconomic uncertainty of Jurado, Ludvigson, and Ng (2015). See Section 4 for details on how each proxy was constructed. In specifications marked (1), the dependent variable has a lead of one period (calendar quarters with respect to the independent variables in specifications marked (2) it leads two periods, and so forth until (4). All specifications include calendar and fiscal quarter dummies, as well as firm fixed effects. All variables are normalized by their sample standard deviation. Standard errors are clustered at the quarter and firm level; t-statistics are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

Table 3: Alternative macroeconomic controls

Table 4
Out-of-sample predictability

Parsimonious model	Larger model	Panel A: Predictive improvement when adding News-based policy uncertainty index to benchmark models			
		(1)	(2)	(3)	(4)
M_0	$M_{0,pu}$	0.00586*** (3.28)	0.00694*** (3.97)	0.00551*** (3.77)	0.00595*** (3.94)
M_1	$M_{1,pu}$	0.00314*** (4.03)	0.00365*** (4.37)	0.00394*** (4.33)	0.00408*** (4.23)
M_2	$M_{2,pu}$	0.000298*** (3.33)	0.000302*** (3.62)	0.000236*** (3.06)	0.000158* (1.67)
M_3	$M_{3,pu}$	0.000193** (2.52)	0.000145** (2.53)	0.000113* (1.93)	0.000140** (2.11)
Parsimonious model	Larger model	Panel B: Predictive improvement when adding controls to random walk model			
		(1)	(2)	(3)	(4)
M_0	$M_{1,pu}$	0.0521*** (17.28)	0.0500*** (18.02)	0.0423*** (18.09)	0.0344*** (16.16)
M_0	$M_{2,pu}$	0.0567*** (17.24)	0.0542*** (17.37)	0.0467*** (16.12)	0.0391*** (15.13)
M_0	$M_{3,pu}$	0.0605*** (16.81)	0.0571*** (16.95)	0.0488*** (15.43)	0.0411*** (13.66)

In this table we use the methodology of Clark and West (2007) to test if the policy uncertainty variable improves the out-of-sample predictive accuracy of the following benchmark models:

$$M_0: INV_{i,t+1} = \alpha_1 + \varepsilon_{i,t+1},$$

$$M_1: INV_{i,t+1} = \alpha_1 + \theta X_{i,t} + \varepsilon_{i,t+1},$$

$$M_2: INV_{i,t+1} = \alpha_1 + \theta X_{i,t} + \gamma IO_{i,t} + \varepsilon_{i,t+1},$$

Table 4: Out-of-sample predictability

6.4 Figure 2: Predicted capital investment levels + Table 5: Cross-sectional heterogeneity in the effect of policy uncertainty

Read:

- Figure 2 compares predicted investment using actual policy uncertainty with counterfactual investment holding policy uncertainty fixed at its 2006Q4 level.
- The investment decline during the crisis period is meaningfully smaller in the counterfactual, suggesting that rising policy uncertainty accounts for a nontrivial part of the investment fall.
- Table 5 shows that policy uncertainty interacts negatively with PPE intensity, low asset redeployability, high sunk costs, durable-goods exposure, and government-spending sensitivity.
- These cross-sectional patterns line up with the real-options prediction that uncertainty matters most when investment cannot easily be reversed or when policy directly affects demand.

Economic message: Figure 2 gives a macro interpretation of the coefficient magnitudes, while Table 5 supplies the mechanism: irreversibility and government exposure amplify the investment response.

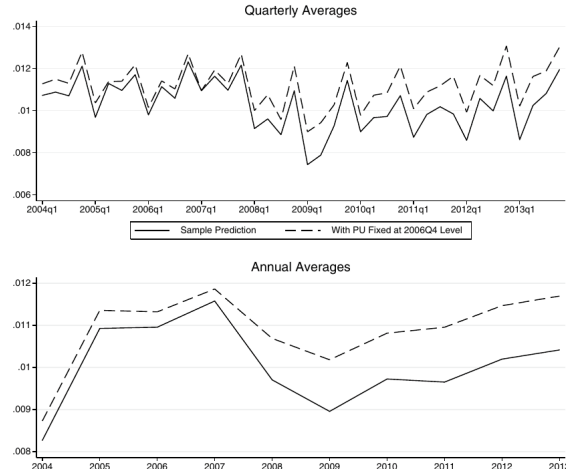


Figure 2: Predicted capital investment levels

Cross-sectional heterogeneity in the effect of policy uncertainty

No controls for interactions with investment opportunities

Controlling for interactions with investment opportunities

Panel A : Interactions with investment irreversibility proxies

Panel A1 : Property, plant, and equipment (PPE)

	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
PU news x PPE	-0.146*** (-15.70)	-0.157*** (-16.82)	-0.145*** (-15.72)	-0.146*** (-15.81)	-0.131*** (-11.57)	-0.128*** (-11.31)	-0.109*** (-9.80)	-0.115*** (-10.45)
Controls	yes	yes	yes	yes	yes	yes	yes	yes

Panel A2 : Asset redeployability (AR)

	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
PU news x AR	-0.0856*** (-10.43)	-0.0757*** (-11.32)	-0.0710*** (-10.71)	-0.0746*** (-11.49)	-0.0621*** (-9.08)	-0.0649*** (-9.04)	-0.0557*** (-7.85)	-0.0620*** (-8.99)
Controls	yes	yes	yes	yes	yes	yes	yes	yes

Panel A3 : High vs. low sunk costs

	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
PU news x Sunk index	-0.0328*** (-4.25)	-0.0481*** (-6.10)	-0.0386*** (-4.68)	-0.0335*** (-4.40)	-0.0396*** (-4.48)	-0.0522*** (-5.70)	-0.0447*** (-4.62)	-0.0430*** (-4.81)
Controls	yes	yes	yes	yes	yes	yes	yes	yes

Panel A4 : Durables vs. nondurables

	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
PU News x Durable index	-0.0589*** (-4.68)	-0.0744*** (-5.76)	-0.0750*** (-5.75)	-0.0833*** (-6.39)	-0.0356*** (-2.23)	-0.0482*** (-3.12)	-0.0393*** (-2.60)	-0.0446*** (-2.94)
Controls	yes	yes	yes	yes	yes	yes	yes	yes

Panel B : Interactions with sensitivity to government spending (SGS)

	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
PU news x SGS	-0.0148* (-1.72)	-0.0346*** (-3.72)	-0.0401*** (-4.34)	-0.0110*** (-3.43)	-0.0194* (-1.89)	-0.0427*** (-3.95)	-0.0438*** (-4.27)	-0.0317*** (-3.10)
Controls	yes	yes	yes	yes	yes	yes	yes	yes

In this table we regress firm-level quarterly investment (CAPX/Lagged Total Assets) on Tobin's q, operating cash flows and sales growth, to which we add the investment irreversibility proxies discussed in Section 3.1 as well as their interaction with the news-based policy uncertainty index from Baker, Bloom, and Davis (2013) (see Equation 3). For expositional clarity, we show only the coefficient estimates of the variables of interest (the interaction between policy uncertainty and the irreversibility proxies). In the last four columns, in all panels, we also control for interactions between all our proxies for investment opportunities from the previous table, and the investment irreversibility proxy used in each panel. Its specifications

Table 5: Cross-sectional heterogeneity in the effect of policy uncertainty

6.5 Figure 3: Effect of policy uncertainty on future investment + Figure 4: Estimated effect of a policy uncertainty shock on aggregate investment

Read:

- Figure 3 shows that investment declines after the onset of a high-policy-uncertainty spell and then gradually recovers.
- The bottom panel of Figure 3 shows that the response is much stronger for firms with low asset redeployability.
- Figure 4 reports a VAR impulse response: a one-unit policy uncertainty shock lowers aggregate investment for several future quarters.
- The response is amplified for low-redeployability firms and close to zero for high-redeployability firms.

Economic message: These figures provide dynamic evidence for precautionary delay. Firms reduce investment when uncertainty rises, and the effect is concentrated where waiting is most valuable.

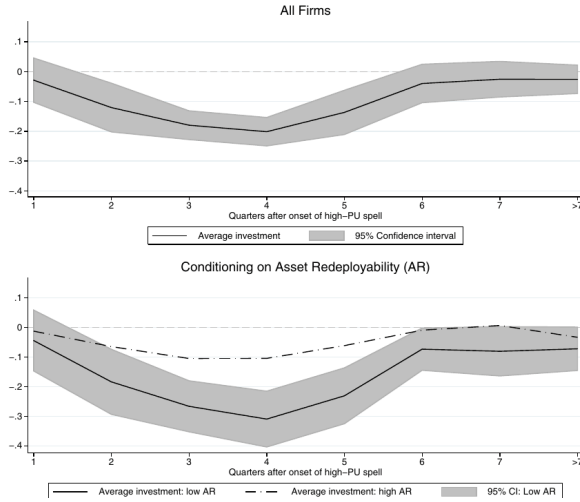


Figure 3: Effect of policy uncertainty on future investment

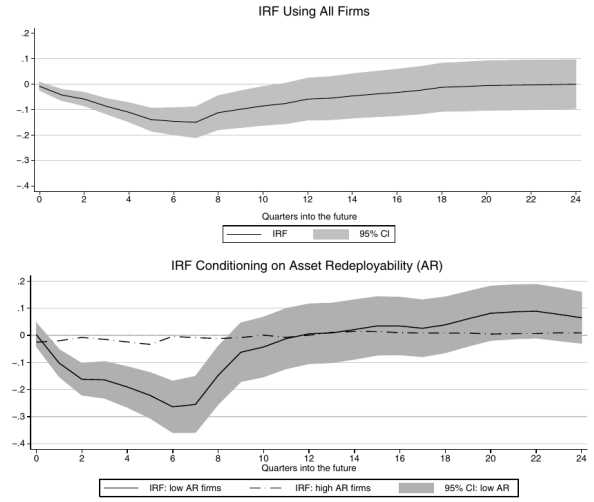


Figure 4: Estimated effect of a policy uncertainty shock on aggregate investment

6.6 Figure 5: Conditional average investment throughout high-policy-uncertainty spells + Table 6: Canadian policy uncertainty, Panel A

Read:

- Figure 5 shows that investment tends to rise later during prolonged high-policy-uncertainty spells, consistent with delayed investment eventually being undertaken.
- The rebound pattern is again different by asset redeployability: firms with high redeployability are less depressed by uncertainty, while low-redeployability firms show a sharper delayed adjustment.
- Table 6 Panel A uses the component of U.S. policy uncertainty orthogonal to Canadian policy uncertainty and other controls.
- The residualized policy uncertainty measure still predicts lower U.S. investment, while it does not predict investment for Canadian firms in the falsification column.

Economic message: Figure 5 strengthens the timing interpretation, and Table 6 reduces the concern that the baseline evidence only captures global macro uncertainty common to North America.

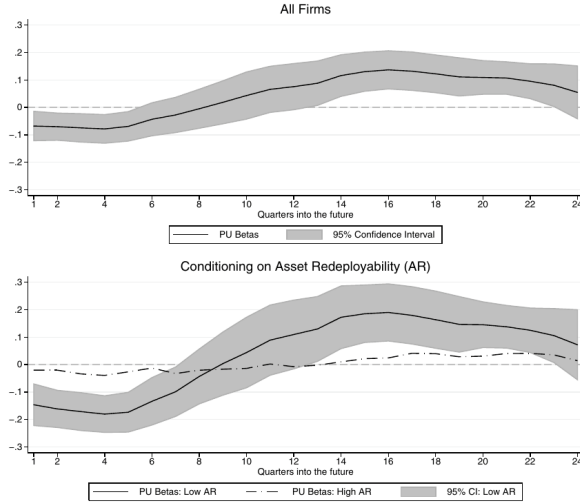


Figure 5: Conditional average investment through high-policy-uncertainty spells

Table 6
Using Canadian policy uncertainty to mitigate endogeneity concerns

Panel A: Average policy uncertainty effect

Dependent variable:	Without first moment controls				With first moment controls				Canada sample
CAPX/Total assets	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(5)
Policy uncertainty (news)	-0.0247*** (-3.38)	-0.0269*** (-2.78)	-0.0280*** (-3.17)	-0.0330*** (-3.87)	-0.0233*** (-3.39)	-0.0256*** (-2.87)	-0.0261*** (-3.49)	-0.0297*** (-4.25)	-0.00940 (-0.36)
Election indicator	0.00195 (0.55)	-0.00048 (-0.07)	-0.00063 (-0.09)	-0.0017 (-1.40)	0.00020 (0.31)	-0.00007 (-0.11)	-0.00069 (-0.74)	-0.0011 (-1.41)	
GDP growth	0.0299*** (4.59)	0.0309*** (5.18)	0.0401*** (5.52)	0.0491*** (6.38)	0.0102 (1.43)	0.0134* (1.71)	0.0114 (1.53)	0.0127* (1.75)	
Expected GDP growth					0.00114 (0.15)	0.00061 (0.83)	0.00079 (1.12)	0.0014 (1.27)	
Leading economic index					-0.00125 (-0.15)	0.007 (1.20)	0.0085*** (2.81)	0.0244*** (3.75)	
Consumer confidence					0.0591*** (7.88)	0.0569*** (7.36)	0.0550*** (8.26)	0.0491*** (6.46)	
Controls	yes	yes	yes	yes	yes	yes	yes	yes	yes
N	342,950	333,001	323,863	316,025	342,950	333,001	323,863	316,025	12,144
R-sq	0.045	0.038	0.032	0.027	0.049	0.043	0.039	0.033	0.044
Firm fixed effects	yes	yes	yes	yes	yes	yes	yes	yes	yes
Quarter dummies	yes	yes	yes	yes	yes	yes	yes	yes	No
Clustering	Cluster bootstrapped standard errors using firm and quarter/year clusters								

In this table we replicate our main results in Table 4 using as our policy uncertainty variable the OLS residuals obtained from regressing the news-based policy uncertainty index in the United States on the news-based policy uncertainty index in Canada, as well as on cross-sectional averages of the control variables used in Table 4. Panel A presents the average effect of this measure of policy uncertainty on capital investments in the United States. Panel B shows interaction effects of this policy uncertainty measure with investment irreversibility (a replication of the last four columns in Table 4). For expositional purposes, the table shows only the variable of interest from each regression. In specifications marked (1), the dependent variable has a lead of one period (calendar quarter) with respect to the independent variables in specifications marked (2) to lead two periods, and so forth until (4). We bootstrap the standard errors using a series of cluster bootstraps as in Cameron, Gelbach, and Miller (2013). t-statistics are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

Table 6: Using Canadian policy uncertainty, Panel A

6.7 Table 6: Canadian policy uncertainty, Panels B–C + Table 7: Instrumental variables

Read:

- Table 6 Panels B–C repeat the heterogeneity tests using residual policy uncertainty. The interaction effects with PPE, asset redeployability, sunk costs, durables, and government-spending sensitivity remain negative.
- Table 7 uses political polarization as an instrument for policy uncertainty.
- Panel A continues to show a negative average effect of instrumented policy uncertainty on capital investment.
- Panels B–C show that the IV strategy also preserves the main heterogeneity results.

Economic message: The main mechanism survives two endogeneity checks: a cross-country residual strategy and an instrument based on political polarization.

Table 6
Using Canadian policy uncertainty to mitigate endogeneity concerns (continued)

Panel B: Investment irreversibility				
Panel B1 : Property, plant, and equipment (PPE)				
	(1)	(2)	(3)	(4)
PU news x PPE	-0.0219*** (-3.06)	-0.0299*** (-4.04)	-0.0256*** (-3.45)	-0.0238*** (-3.82)
Controls	yes	yes	yes	yes
Panel B2 : Asset redeployability (AR)				
	(1)	(2)	(3)	(4)
PU news x Asset redeployability	-0.0138*** (-2.62)	-0.0227*** (-4.51)	-0.0191*** (-3.70)	-0.0177*** (-3.62)
Controls	yes	yes	yes	yes
Panel B3 : High vs. low sunk costs				
	(1)	(2)	(3)	(4)
PU news x Sunk index	-0.0114*** (-2.62)	-0.0162*** (-3.38)	-0.0161*** (-3.77)	-0.0146*** (-3.29)
Controls	yes	yes	yes	yes
Panel B4 : Durables vs. nondurables				
	(1)	(2)	(3)	(4)
PU news x Durable index	-0.00421 (-1.10)	-0.00679** (-2.12)	-0.00594* (-1.71)	-0.00590* (-1.67)
Controls	yes	yes	yes	yes
Panel C : Sensitivity to government spending (SGS)				
	(1)	(2)	(3)	(4)
PU news x Sensitivity to gov. spending	-0.00835* (-1.70)	-0.0162*** (-3.84)	-0.0174*** (-4.54)	-0.0150*** (-3.87)
Controls	yes	yes	yes	yes

Table 6: Using Canadian policy uncertainty, Panels B–C

Table 7
Using instrumental variables to mitigate endogeneity concerns

Dependent variable: CAPX/Total assets		Panel A: Average policy uncertainty effect			
	(1)	(2)	(3)	(4)	
Policy uncertainty (news)	-0.0327*** (-2.99)	-0.0314*** (-2.81)	-0.0264** (-2.23)	-0.0223* (-1.88)	
Controls	yes	yes	yes	yes	
Firm fixed effects	yes	yes	yes	yes	
Quarter dummies	yes	yes	yes	yes	
Clustering	Cluster-bootstrapped standard errors using firm and quarter clusters				
Dependent variable: CAPX/Total assets		Panel B: Investment irreversibility			
		Panel B1 : Property, plant, and equipment (PPE)			
	(1)	(2)	(3)	(4)	
PU news x PPE	-0.126*** (-9.30)	-0.135*** (-10.99)	-0.133*** (-9.61)	-0.129*** (-10.69)	
Controls	yes	yes	yes	yes	
		Panel B2 : Asset redeployability (AR)			
	(1)	(2)	(3)	(4)	
PU news x Asset redeployability	-0.0469*** (-11.69)	-0.0515*** (-14.62)	-0.0514*** (-13.58)	-0.0498*** (-13.46)	
Controls	yes	yes	yes	yes	
		Panel B3 : High vs. low sunk costs			
	(1)	(2)	(3)	(4)	
PU news x Sunk index	-0.210*** (-3.86)	-0.213*** (-3.52)	-0.201*** (-3.49)	-0.194*** (-3.39)	
Controls	yes	yes	yes	yes	
		Panel B4 : Durables vs. nondurables			
	(1)	(2)	(3)	(4)	
PU news x Durable index	-0.185*** (-4.05)	-0.194*** (-5.28)	-0.161*** (-3.97)	-0.153*** (-3.92)	
Controls	yes	yes	yes	yes	
		Panel C : Sensitivity to government spending (SGS)			
	(1)	(2)	(3)	(4)	
PU news x Sensitivity to gov. spending	-0.170*** (-3.76)	-0.170*** (-4.29)	-0.148*** (-4.52)	-0.0867** (-2.53)	

Table 7: Using instrumental variables to mitigate endogeneity concerns

7 Short summary

- The paper studies the relationship between policy uncertainty and corporate capital investment.
- The baseline policy uncertainty variable is the Baker–Bloom–Davis news-based index.
- Higher policy uncertainty predicts lower investment over the next one to four quarters.
- The result survives controls for Tobin’s q , cash flow, sales growth, GDP growth, election years, expected GDP growth, leading indicators, confidence, forecast dispersion, volatility, and macro uncertainty.
- Policy uncertainty improves out-of-sample prediction of future investment.
- The effect is stronger for firms with more irreversible investment and for firms more sensitive to government spending.
- The dynamic evidence shows a decline in investment after uncertainty rises and a later rebound, especially when investment is more reversible.
- Canadian residual policy uncertainty and political polarization IV tests reduce endogeneity concerns.
- The central contribution is to connect policy uncertainty, real-options investment delay, and firm-level investment heterogeneity.

8 Conclusion

8.1 Core conclusion

Gulen and Ion show that policy uncertainty is a quantitatively important predictor of corporate investment. When uncertainty about future policy and regulation rises, firms reduce capital expenditures, and the effect is strongest for firms whose investment is hardest to reverse.

8.2 Incremental lesson relative to standard investment models

Standard investment models emphasize Tobin’s q , cash flow, and sales growth. This paper shows that a macro-political state variable adds explanatory and forecasting power. The effect is not just an average time-series relation; it is strongest where real-options theory predicts it should be strongest.

8.3 Mechanism

The evidence points toward precautionary delay. When policy uncertainty rises, managers face a higher option value of waiting. For firms with specialized assets, high sunk costs, or low redeployability, the downside of committing capital before the policy environment is known is larger. These firms cut investment more and recover more slowly.

8.4 Identification and limitations

The paper’s main empirical challenge is that policy uncertainty rises during bad macroeconomic periods. The authors address this with macro controls, out-of-sample forecasting tests, Canadian residualization, and political-polarization IV analysis. These tests strengthen the case, but the evidence is still best read as quasi-causal rather than fully experimental.

8.5 Broader implication

The paper gives a corporate-finance channel through which political systems affect real activity. Unpredictable policy can depress investment even when firms are not financially constrained. The results suggest that policy stability has value not only because it changes expected cash flows, but because it reduces the incentive to wait.

8.6 Takeaway

Policy uncertainty matters because investment is irreversible. The paper's key empirical contribution is showing that policy uncertainty predicts investment most strongly for firms that real-options theory identifies as most exposed to delay incentives.